V

Valves (*see* Control Systems)

Vanes (*see* Metallurgy)

Vaporizers; Vaporizer Applications*

Applications are easier to discuss with specific reference to certain models. (See Figs. V-1 through V-6.)

Types of Vaporizers

- Vertical vaporizer ("vertical bayonet"): It is widely used for chlorine, ammonia, propane, methanol, sulfur dioxide, etc. Sizes range from 50,000 to 15,000,000 Btu/h (12,500 to 3,750,000 kcal/h). Very compact, high productivity, easily combined with built-in superheater with common control. Many heating media can be used, including steam, hot water, and heat transfer fluids such as Dowtherm, Therminol, etc. Electric heated vaporizers also available. Small footprint. (See Fig. V-7.)
- Indirect fluid heater: Very useful for high-pressure or corrosive fluids where special metallurgy (i.e., corrosion-resistant metals) can be used in smaller, less costly containment than traditional shellside boiling. Heating medium (steam/dowtherm/electric, etc.) heats an intermediate bath of water/NH3/Therminol or similar heat-transfer fluid that then heats a second coil at much lower cost than shellside heating or boiling. (See Fig. V-8.)
- Tubular low-temperature vaporizers/superheaters: Combination large flow rate liquid heatup and subsequent boiling or superheating of mixed fluids with diverse boiling points. Needs special stress analysis and mechanical design. Can preheat, boil, and superheat in same vessel. (See Fig. V-9.)
- Impedance electric heaters: Electric heater for process fluids. Lowest cost heater for life of equipment. Easily cleanable, very safe, very long life, simple maintenance, good for high temperature boiling/heat to 2000°F (1093°C), very useful for remote locations of corrosive fluids or gases. Electric current flows through the containment tube and generates heat that is transferred to the fluid. (See Fig. V-10.)
- Electric resistance vaporizers: Classic reboilers or submerged resistance heating elements in normal shells for pool boiling duty. Useful for low to medium capacity loads and more common metals of construction. Can have combined superheat coils in the same containment vessel. (See Fig. V-11.)
- Vaporizers with controls: Indirect fluid electric vaporizer with controls mounted. Very high-pressure heater of corrosive fluid. Fluid side 3175 psi (223 kg/cm^2 or 216 ATM) design pressure. (See Fig. V-12.)

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* Source: Armstrong Engineering Associates, USA. Adapted with permission.



FIG. V-1 Very large low-temperature vaporizer/superheater, steam heated. Unit size 132 in (3353 mm) in diameter by 42 ft (13 cm) long. Duty to boil and superheat very low-temperature organic liquid. (Source: Armstrong Engineering Associates.)

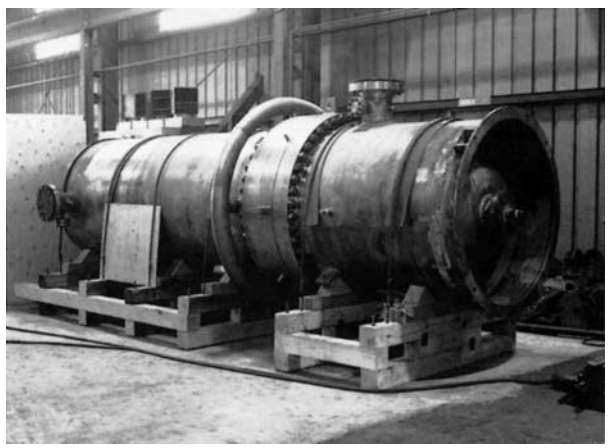


FIG. V-2 One of six all-stainless-steel vertical vaporizers for vertical location above tower in Mideastern refinery. Unit size 59 in (1500 mm) in diameter by 20 ft (6100 mm) high. (Source: Armstrong Engineering Associates.)

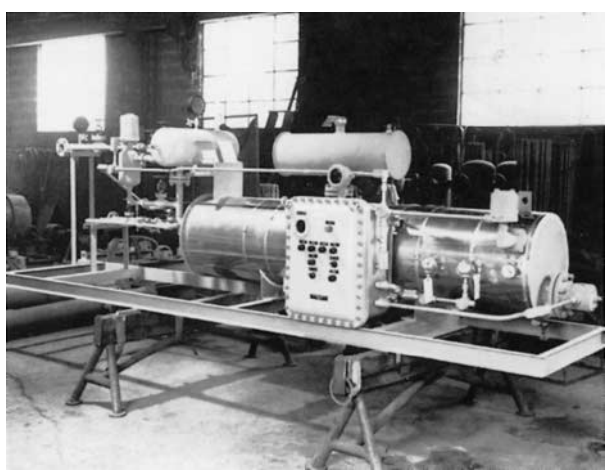


FIG. V-3 Indirect fluid electric heater insulated and mounted on skid with all controls in place. One of several at the same site in South America, vaporizing organic fluids. (Source: Armstrong Engineering Associates.)

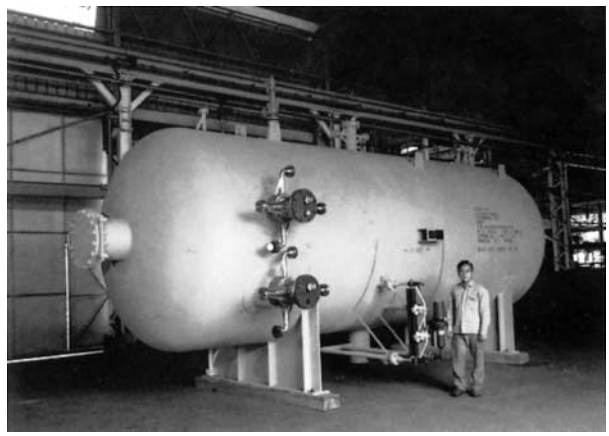


FIG. V-4 Large-sized ethylene vaporizer (steam in shell) with entering temperature of ethylene –67°F (–55°C) installed in India. Steam-heated steel shell with stainless steel heat-transfer surface. (Source: Armstrong Engineering Associates.)



FIG. V-5 Seven large hydrocarbon fluid vaporizers for chemical plant in Middle East. Unit size 36 in (914 mm) in diameter, overall height 135 in (3430 mm). (Source: Armstrong Engineering Associates.)

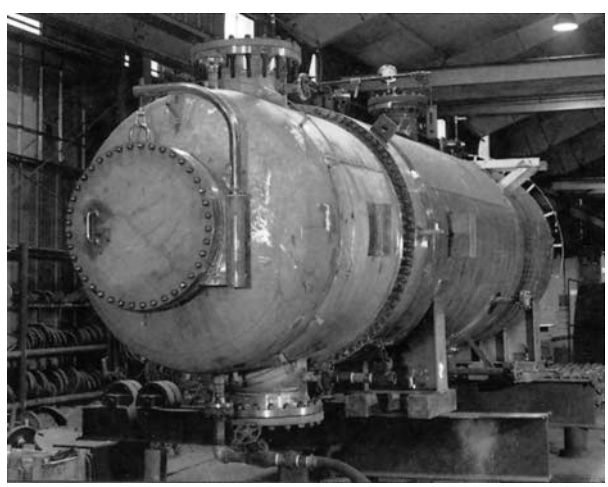


FIG. V-6 In-tube cryogenic vaporizer/superheater with steam in shell and mixed hydrocarbons on tubeside. Steel shell with balance stainless steel for U.S. installation. (Source: Armstrong Engineering Associates.)

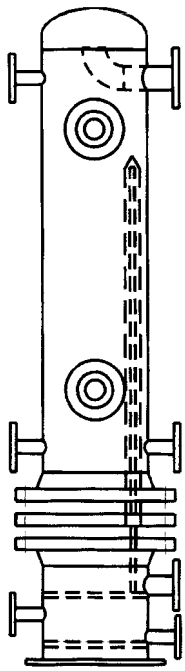


FIG. V-7 Vertical bayonet. (Source: Armstrong Engineering Associates.)

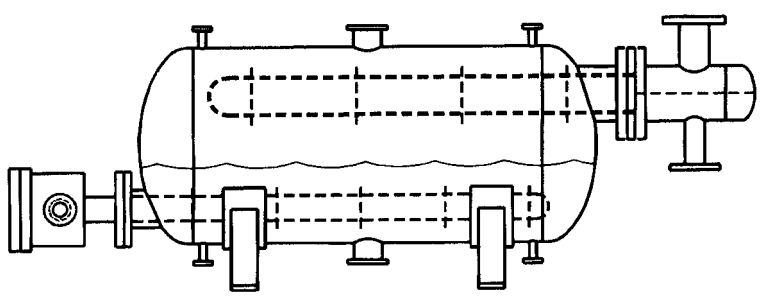


FIG. V-8 Indirect fluid heaters. (Source: Armstrong Engineering Associates.)

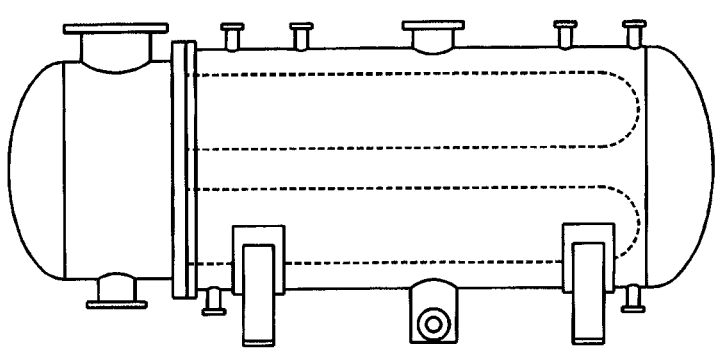


FIG. V-9 Tubular low-temperature vaporizers/superheaters. (Source: Armstrong Engineering Associates.)

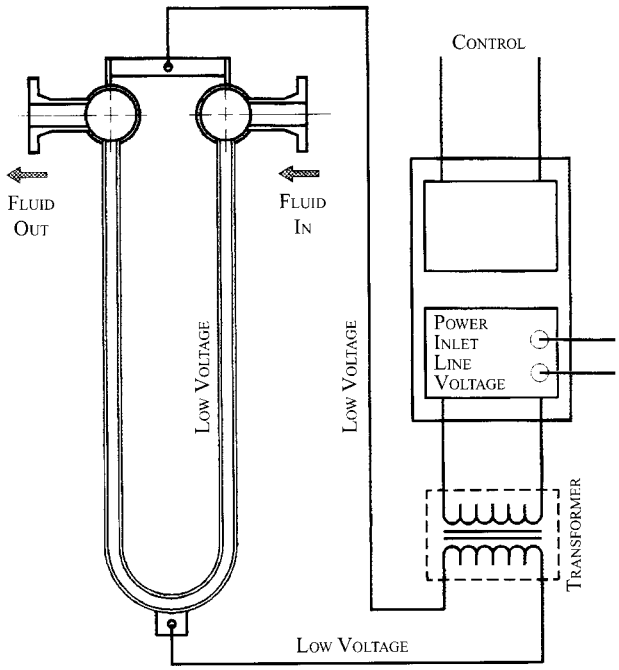


FIG. V-10 Impedance (electric) heaters. (Source: Armstrong Engineering Associates.)

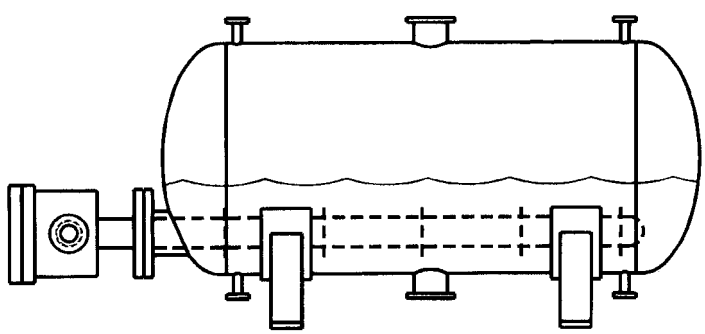


FIG. V-11 Electric resistance vaporizers. (Source: Armstrong Engineering Associates.)

- Electrical radiant furnaces: Radiant furnaces for high-temperature boiling levels of corrosive fluids or heating up to very high exit temperatures above fluid heating media capability [i.e., 2000°F (1093°C)]. Also for very high-pressure or corrosive fluids. Sizes from 12 to 50 ft (3.6 to 15.3 m) high. Can be very high capacity [some about 15,000 kW (12,900,000 kcal/h)] near nuclear site. (See Fig. V-13.)
- Cryogenic vaporizer: For boiling very low temperatures [–327°F (–200°C)]. Flare drum duty, to meet a few second startup emergency. Heating medium in shell and boiling fluid inside the tubes. Must be able to cope with thermal expansion and adjustments in a few seconds without damaging stresses. Also must avoid metallurgical problems including fatigue (cycling) for duties at high pressure such as ethylene, etc. Avoid freeze-up problems and heat up the fluid to required exit temperatures with no accompanying freezeup problems. Also, used to heat subzero fluids being distributed on service grids to multiple users and cold fluids

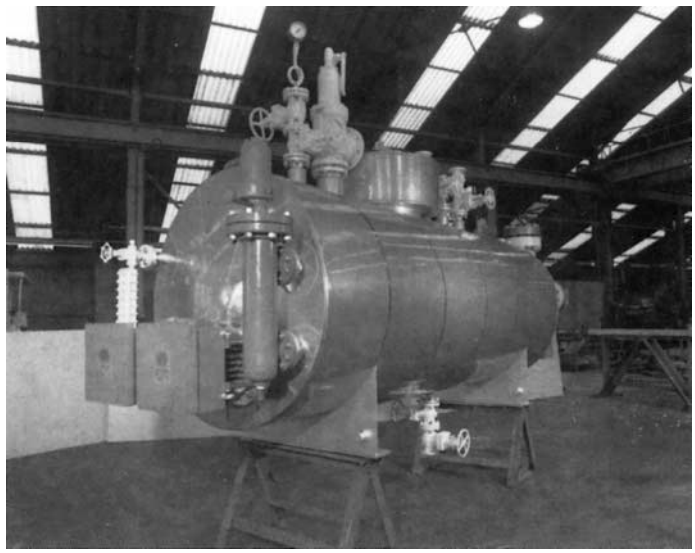


FIG. V-12 Vaporizers with controls or on skids with controls mounted. (Source: Armstrong Engineering Associates.)

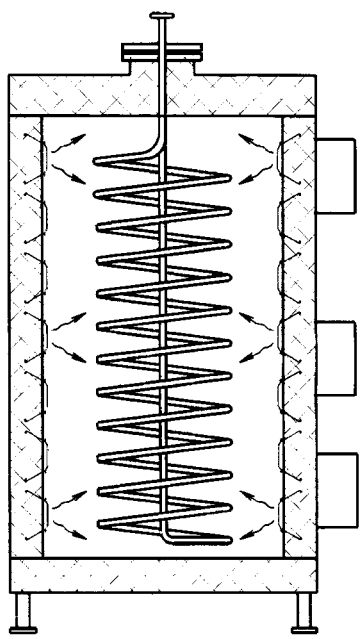


FIG. V-13 Electrical radiant furnaces. (Source: Armstrong Engineering Associates.)

from ships or rail cars needing heatup to avoid fracture of steel or other nonductile piping systems of user. Sizes can be up to 12 ft (3.6 m) in diameter and 40 ft (12 m) long. Shells often steel with tubes of stainless steels 304/316, etc. (See Fig. V-14.)

Vaporizer Specifications and Process Parameters

See Tables V-1 and V-2.

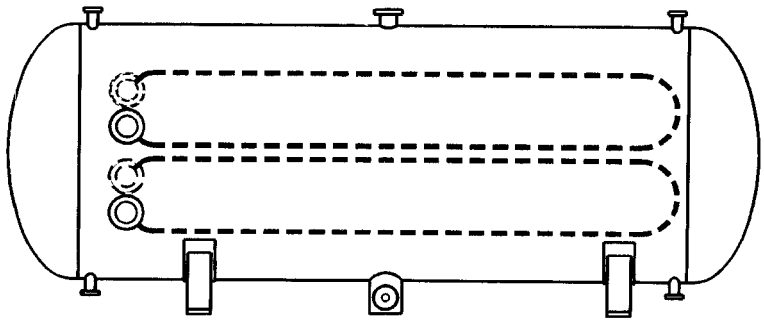


FIG. V-14 Extremely low temperature. (Source: Armstrong Engineering Associates.)

TABLE V-1 Useful Conversion Factors

	U.S. or British Units	SI and Metric Units	Multiplier to Convert	
			SI/Metric to British	British to SI/Metric
Heat duty	Btu/h	Watts	3.4144	0.29288
	Btu/h	kcal/h	3.9683	0.2520
Pressure	psi	kg/cm^2	14.223	0.0703
	psi	Bars	14.504	0.0689
	psi	Pa	$1.4504(10)^{-4}$	$6.8948(10)^{3}$
Velocity	ft/sec	m/s	3.2808	0.3048
Volumetric flow rate	gal/min (US)	m^3/s	$1.585(10)^{4}$	$6.309(10)^{-5}$
Mass flow rate	lb/hr	kg/s	$7.936(10)^{3}$	$1.260(10)^{-4}$
Density	lb/ft^3	kg/m^3	0.06242	16.018
Heat capacity	Btu/lb—F	J/kg—K	$2.3901(10)^{-4}$	$4.1840(10)^{3}$
Enthalpy	Btu/lb	J/kg	$4.302(10)^{-4}$	$2.324(10)^{3}$
	Btu/lb	kcal/kg	1.8000	0.55556
Viscosity	$lb_m/hr—ft$	cp	2.419	0.4134
Thermal conductivity	Btu/h—ft—F	W/m—K	0.57818	1.7296
	Btu/h—ft—F	kcal/h—m—C	0.67197	1.4882
Heat flux	$Btu/h—ft^2$	W/m^2	0.3172	3.1525
	$Btu/h—ft^2$	$kcal/h—m^2$	0.3686	2.7125
Heat-transfer coefficient	$Btu/h—ft^2—F$	$kcal/h—m^2—C$	0.2048	4.8824
	$Btu/h—ft^2—F$	$W/m^2—K$	0.17623	5.6745

Terminology: cm = centimeter, C = degrees Celsius, cp = centipoise, F = degrees Fahrenheit, ft = feet, gal = U.S. gallons, h = hour, J = joules, K = degrees Kelvin, kcal = kilocalories, kg = kilograms, m = meter, min = minute, Pa = pascals, s = seconds, W = watts.

Example: To convert heat duty in Btu/h to kcal/h, multiply Btu/h × 0.252, e.g., 15,000,000 Btu/h × 0.252 = 3,780,000 kcal/h. To convert velocity in ft/s to m/s, multiply by 0.3048: Velocity of 5 ft/s × 0.3048 = 1.524 m/s.

Vertical bayonet vaporizers

See Fig. V-15 and Table V-3.

Specifications

Shell. Ruggedly fabricated welded steel. Shells 24 in (610 mm) and below are made of SA-106 Gr.B pipe. Larger shells are welded of steel plate of SA-516 Gr.70 normally. Tubesheets are normally of SA-516 Gr.70 material, but are also available in stainlesses, nickel alloys, Hastelloy, etc.

Tube bundle. Removable on all sizes if required. Standard design of size "A" and "B" units may not have a removable tube bundle. Tubes are normally 1 in (25.4 mm)

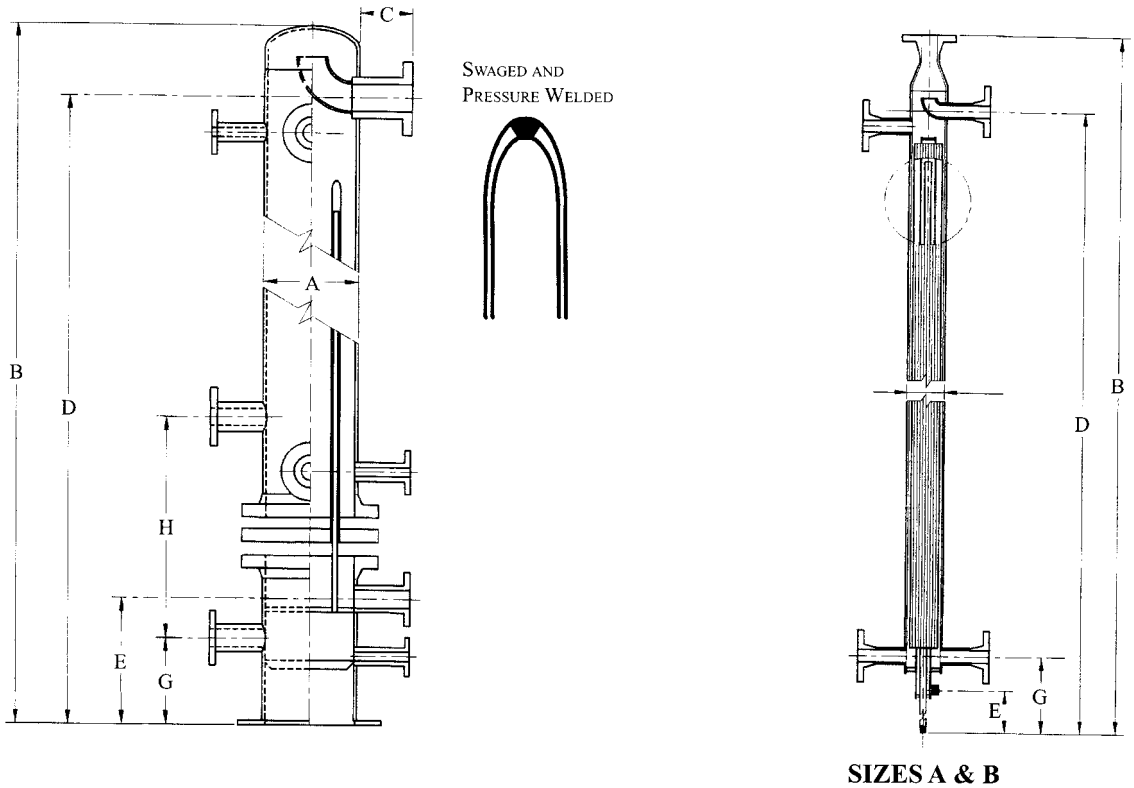


FIG. V-15 Vertical bayonet vaporizer dimensions. (Source: Armstrong Engineering Associates.)

TABLE V-2 Comparison of American, British, German, and Japanese Material Specifications

Material	American (ASME/ASTM)	German (DIN)	British (BS)	Japanese (JIS)
Plates	SA 516 Gr 60–70	A St 45–52 DIN 17135	BS 1501-224-490	JIS G 3118 SGV 49
	SA 515 Gr 60	H II DIN 17155	BS 1501-161-430	JIS G 3103 SB 42
Pipes	SA 53 Gr B	ST 45 DIN 1629	BS 3601 HFS-430	STPG42-G3454
(seamless)	SA 106 Gr A	ST 35.8 DIN 17175	BS 3602 HPS-360	STPT 38-G3456
	SA 106 Gr B	ST 45.8 DIN 17175	BS 3602 HFS-430	STPT42-G3456
Tubes	SA 214	ST 37.2 DIN 1626	BS 3059 ERW-320	STB 35E-G3461
	SA 179	ST 35.8 DIN 17175	BS 3059 CDS-320	STB 30S-G3461
Forgings	SA 105	C22.8Vd TUEV 350/3	BS 1503-221-490	S25C-G4051
Studs	SA 193 GR B7	21CrMoV57 DIN 17240	BS 4882 Gr B7	SNB7-G4107
Nuts	SA 194 GR 2H	24CrMo5 DIN 17240	BS 4882 Gr 2H	S45C-G4051

O.D., 0.083 in (2.1 mm) Bwg. but can be changed to meet customer specifications. Top ends of tubes are securely welded shut on all units. Tubes on sizes A and B normally have external longitudinal fins in contact with liquid being vaporized, multiplying the external surface about eight times, but can also be supplied with bare internal heating tubes.

Tubes are welded to the tubesheet and then rolled and expanded for additional holding power. Rolled joints alone are not sufficient for extended periods of service. For special services, these tubes can be of steel, stainless steel, or other materials. Bayonet tubes are roller expanded into lower tube plate.

TABLE V-3 Approximate Dimensions* (for Steam-Heated Vessels) (Dimensions Are in Millimeters)

Size	Base Plate Dia.	Shell O.D. "A"	Overall Height "B"	Outlet Projection "C"	Height of Outlet "D"	Cond. Height "E"	Inlet Height "G"	Outlet Height "H"	Steam Inlet	Cond. Outlet	Liquid Inlet	Vapor Outlet	Float & Safety
A	N/A	89	1867	152	1715	64	165	n/a	19	13	25	38	38
B	N/A	114	1981	152	1880	127	229	n/a	13	13	38	38	38
BT	203	114	1930	152	1753	191	127	343	25	25	38	38	38
C	305	168	2134	152	1930	337	270	378	38	38	51	51	51
D	305	219	2134	152	1930	356	270	378	38	38	51	51	51
E	413	273	2134	152	1930	356	270	378	51	51	64	64	51
F	413	324	2159	152	1930	330	229	419	76	76	76	76	51
G	508	406	2134	152	1930	311	203	479	102	51	102	102	51
H	660	508	2337	152	2070	368	270	498	102	51	102	102	51
I	762	610	2337	152	2057	368	270	537	102	51	102	102	51
J	889	762	2946	203	2578	648	508	737	152	76	102	152	51
K	1041	914	3200	203	2718	737	559	864	203	102	102	203	51
L	1194	1067	3404	203	2832	889	660	826	254	102	152	254	51
M	1346	1219	3556	203	2972	914	660	864	305	152	152	305	51

* Dimensions and outlet sizes may be varied to suit individual job conditions. Gauge connection is 3/4 in. All nozzles $2^1/_2$ in and over are flanged.

NOTE: Outlets with screwed connections also available upon request (at lower cost).

TABLE V-4 Approximate Steam Consumption

U.S. units: One Pound of Steam per Hour Will Vaporize		Metric units: One kg/h Steam Will Vaporize
1.0 U.S. gallon/h	Propane Butane	8.4 L/h
2 lb/h	Ammonia	2 kg/h
8 lb/h	Chlorine	8 kg/h
6 lb/h	Sulfur dioxide	6 kg/h

Design working pressure. On the process side, normally 250 psi (17.6 kg/cm^2). In steam or hot water space, 100 psi (7 kg/cm^2) (higher pressures available if needed). All vaporizers built in the U.S. are designed, inspected, and National Board stamped in accordance with the ASME Code. Vaporizers built outside the U.S. may be supplied per ASME, TUV, Stoomwezen or other local codes as required.

When operating pressure goes below 25 psi on propane or butane, check with the factory to avoid difficulty from pressure drop through nozzles. (See Table V-4.)

The 1-in-O.D. (25.4 mm) × 0.083 in (2.1 mm) wall tubes, seal-welded and rolled, give more clearance for condensate and steam. Thicker tube used in design adds to the life of the bundle. The seal-welding, and rolling gives strength needed against the fairly rapid variations in pressure and temperature encountered under some conditions. To avoid a loss of process fluid through leakage (and the peril of a potential explosion), the OEM seal-welds the tubes to the tubesheet.

See Figs. V-16 through V-18.

Design features

The tubes are free to expand or contract. Since the tubes are only secured at the bottom end, there is no tendency for the tubes to flex or twist from temperature stress. This is a marked advantage over units with tubesheets at both ends, where repeated temperature stress may cause failure at the tube end.



FIG. V-16 16-in-diameter (406-mm) chlorine vaporizer. (Source: Armstrong Engineering Associates.)



FIG. V-17 Four typical vertical vaporizers for large Mideastern refiner. Note the inlet belts on three of the vaporizers, often used to improve shell side distribution for improved boiling and excessive tube impingement. (Source: Armstrong Engineering Associates.)

Bottom steam feed protects against freezeup. The condensate is constantly warmed by incoming hot steam or hot water (if that is the heating medium). Even though the vaporizing temperature in the shell falls below freezing temperature, the condensate does not run the risk of freezing with consequent bursting of a tube. For boiling below 25°F (–4°C), consult the information source.

The tube bundle is removable and can be replaced in the field. It is no longer necessary to remove the whole unit in the event that the tubes begin to corrode out. A replacement bundle can be bought and installed in the field with a minimum amount of downtime.

How the vaporizer works—unlimited built-in turndown

1. The vaporizer takes its feed from a storage tank or process plant output and the boiling liquid rises in the shell until the vapor outlet generated by the load is matched by the heat transfer to the submerged surface level at the time. At that point, it stabilizes and continues to boil at that level until the load changes. If the load rises, the level of the fluid goes up to give the added output needed. If

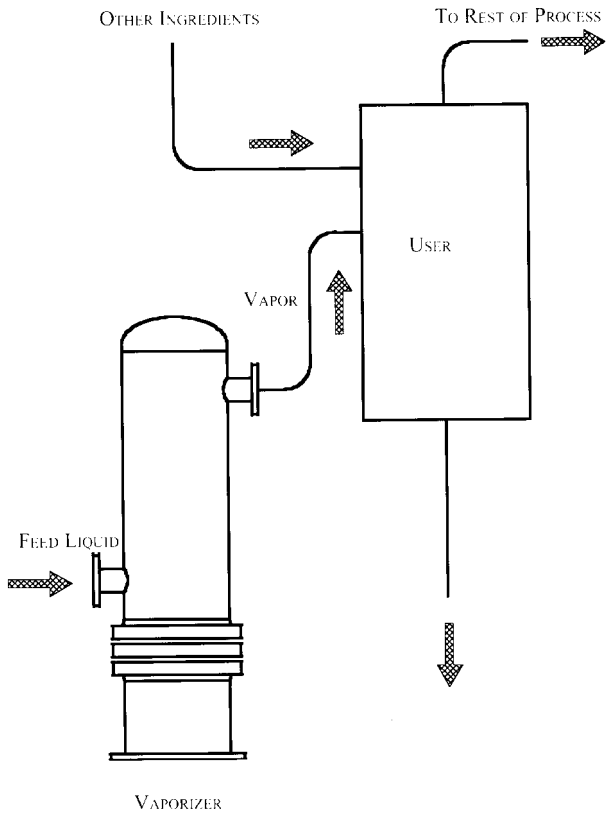


FIG. V-18 Vaporizer in process flow. (Source: Armstrong Engineering Associates.)

the load drops, the fluid level in the vaporizer drops until the output matches demand. This automatic turndown applies to any operating level in the vertical vaporizer. No special turndown control is needed.

2. It is easy to include a superheat section by adding height to the bundle (added surface) sufficient to achieve the desired superheat. This is impossible in either jacketed shell or reboiler-type vaporizers without the addition of a separate superheating element at substantial added cost.
3. These units protect against freezeup when boiling near the freezing point of the steam of other heating medium. The vaporizer can operate with boiling temperatures somewhat below the freezing point. The bottom steam feed protects the tubes against freezing so that the vaporizer can operate at boiling temperatures below the freezing point of the heating medium condensate. Consult factory for specific design figures in such cases.
4. The hold-up volume of process fluid is well below that in other types of vaporizers.
5. The footprint of the vaporizer is less than any horizontal unit. It is also normally lower than any other type of vertical in-tube or jacketed unit because of the greater output of the vaporizer.
6. The vaporizer is a standardized design and preliminary layout drawings are available early to enable plant layout to proceed quickly.
7. The tube bundle is removable and can easily be replaced or changed in the field.

8. The tubes are secured only at one end and are free to expand or contract so there is no thermal stress originating due to temperature variations in the bundle.
9. Code approval is normally easy since almost all code supervision agencies in the world have experienced submissions of vaporizers in past years.

Freeze-up protection with bayonet-type vaporizers

Controls and recommendations. During normal operation, the vertical bayonet design is excellent for vaporizing fluids at temperatures of 32°F (0°C) or several degrees lower. The leaving condensate is constantly warmed by incoming hot steam. The following recommendations are based on operating experience of vaporizers for propane, ammonia, chlorine, etc.

Precautions against freezing of steam condensate

Steam failure. The steam controls should be arranged such that the steam cannot be shut off at any time when cold process liquid can be in the shell at or below 32°F (0°C) and the operating instructions to personnel should stress this fact.

As an example, if a thermostatic steam valve or similar control is used in the inlet steam line, it should be limited in such a way that it cannot shut off completely when the process fluid in the shell is below 32°F (0°C). A hand valve in the steam line as a bypass around the control valve may be used to provide a positive steam supply.

Startup procedure would be to first establish steam supply to the unit before permitting cold process liquid to enter the shell, and shutdown procedure would be to first stop the process fluid flow before stopping the steam. If there is a failure of the steam supply, some precaution is desirable to stop the process fluid flow and to immediately remove the cold process fluid from the shell.

Suggestions would include a temperature control switch in the condensate line to sound an alarm and/or stop process fluid flow. A control indicating steam pressure failure may also be used.

Condensate backup. The steam and condensate lines must be free draining. In the case of a condensate return line to the boiler, care must be taken that the steam pressure is high enough to avoid a static head in the condensate line, which may result in backing up of condensate into the steam space of the vaporizer. This condensate may then freeze if cold process fluid is present in the shell.

Steam trap. The steam trap must be adequately sized to avoid backup. Also, a trap with minimum holdup of condensate is preferable. If the steam fails, condensate will re-evaporate and return to the tubes, so an absolute minimum condensate volume in the trap is desirable. Thermostatic traps have proven satisfactory for many applications.

Separate trap on steam chamber. A separate trap is recommended to carry away condensate that forms in the steam feed line and in the steam chamber.

Trap not too high. The trap on the main condensate outlet should be installed enough below the vaporizer condensate outlet connection to avoid backing up of condensate inside the vaporizer due to equalizing loads.

Strainers on traps. The traps should be equipped with strainers to ensure foreign materials will not plug the trap.

Positive steam pressure. The steam should be operated at a high enough pressure to overcome any pressure loss in lines, valves, fittings, etc., and to ensure operation

of the steam trap. Typically a pressure of 5 to 15 psig (0.35 to 1.05 kg/cm^2) is used as a minimum.

Steam trap stoppage is, arguably, the single most prevalent cause of freeze-up in vaporizers. For critical installations, duplicate traps may be installed in parallel.

Superheating outlet vapor methods and reasons

Basic design of vaporizers includes:

- Sensible heat to warm up the liquid from storage temperature to the boiling temperature in the vaporizer.
- Latent heat to boil the liquid at vaporizer temperature and pressure.
- Superheat required to heat vapor from saturation temperature to some desired gas outlet temperature.

The vaporizer usually has enough surface, figured to operate below the liquid level, to preheat the liquid and boil it.

Any surface required to superheat must be above the boiling area.

Three basic approaches to superheat are used:

1. Extension of tube bundle above liquid boiling level to add superheating surface.
2. A completely separate external superheater can be used.
3. For many fluids, reducing the pressure of discharged saturated vapor will produce some superheat. This method may invite surging. Consult factory.

Extension of the tube bundle. This usually requires more surface than #2 (above) as the vapor velocity is lower. Control of the gas outlet temperature is somewhat difficult since there is only one steam supply. However, by maintaining the boiling in the vaporizer at a fixed pressure, and setting a steam control to a fixed gas outlet temperature, control is possible.

This method is often somewhat more compact and also less costly if the amount of superheat is not too great.

Reasons for superheating

- Superheat may be required where outlet vapor lines are long, uninsulated, or exposed to low temperatures, so that recondensation could take place. Initial superheat allows for line temperature losses and the vapor can be delivered intact at the pipeline outlet end.
- Some controls contain elements subject to freeze-up or damage at temperatures below 32°F (0°C). Superheat of the vapor will avoid this danger.
- Controlled superheat may often be required for process reasons.

Separate external superheater. This is most desirable in cases where a large amount of superheat is needed. Superheaters are often made with finned tubes, which give a less costly heat exchanger than one with bare tubes for this duty.

Control is simple.

See Figs. V-19 and V-20.

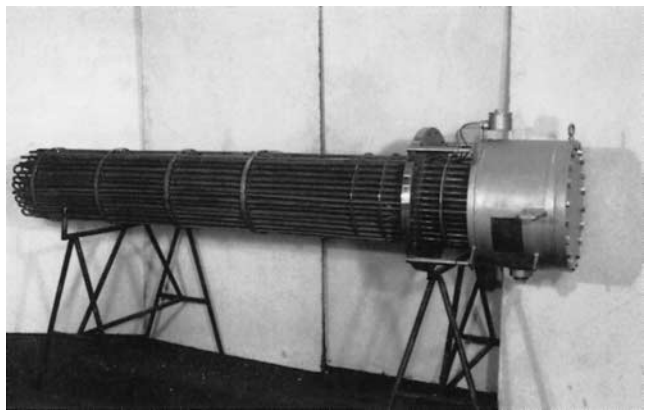


FIG. V-19 Typical electric superheater. (Source: Armstrong Engineering Associates.)

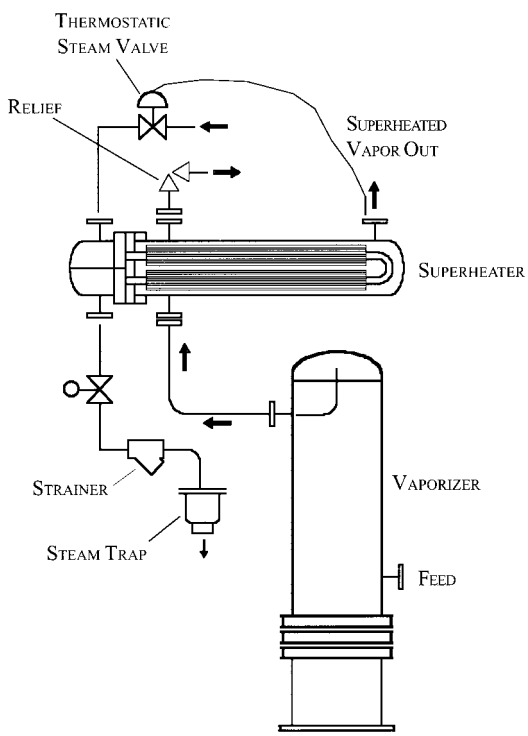


FIG. V-20 The above control hookup shows a typical external superheater. The advantage of the separate superheater is twofold. First, by having a separate steam feed on the superheater controlled by gas outlet temperature, definite temperature control can be gained. Second, the superheating surface is much cheaper as finned tubing than it would be in adding bare tube surface to the vaporizer bundle for boiling. Every installation using this system has been quite successful, although there are a number of jobs on which superheat was obtained merely by incomplete immersion of the bundle. This does take quite a bit more surface; however, each job should be individually calculated. (Source: Armstrong Engineering Associates.)

Some vaporizer applications

Liquid Vaporized	End Product	Typical Process
Acetaldehyde	N-Butyl Alcohol	Acetaldehyde based
Acetic acid	Acetic anhydride	Acetylene
	Acetaldehyde	Acetic acid based
	Vinyl acetate	Acetylene based
Ammonia	Acryonitrile	Sohio
	Ethanolamines	Ethylene oxide based
	Fertilizers	Various
	Nitric Acid	Ammonia based
Benzene	Maleic anhydride	Various
Butadiene	1,4-Dichloro-2-Butylene	Butadiene based
Butane	Acetaldehyde	
	Acetic acid	Oxidation of butane
	Acetone	
	Formaldehyde	
	MEK, Methanol	
	Propanol	
Butylene	Isoprene rubber	Prins
Chlorine	Aldrin, dieldrin	
	Endrin, isodrin	Shell
	Carbon tetrachloride	
	Perchloroethylene	Propane chlorination
	Chlorine dioxide	Various
	Chlorobenzenes	Various
	Chloromethanes	Methane based
	1,4-Dichloro-2-Butylene	Butadiene based
	Ethyl chloride	Ethane, ethylene based
	Ethylene dichloride	Ethylene based
	Ethylene oxide	Epichlorohydrin
	Glycerine	Propylene based
	Phosgene	Carbon monoxide based
	Propylene oxide	Epichlorohydrin
	Titanium dioxide	Various
	Vinylidene chloride	Acetylene based
Ethanol	Acetaldehyde	Ethanol based
Ethylene dighloride	Vinyl chloride	Ethylene dichloride based
Ethylene oxide	Ethanolamines	Ethylene oxide based
Formaldehyde	Isoprene rubber	Prins
Hydrogen chloride	Bisphenol	Various
	Ethyl chloride	Ethylene based
	Methyl chloride	Methanol based
	Neoprene	Acetylene based
	Trichloroethylene	
	Perchloroethylene	Acetylene based
	Vinyl chloride	Wulff
Hydrogen cyanide	Acrylonitrile	Acetylene based
	Acrylonitrile	Ethylene oxide based
Methanol	Dimethyl terephthalate	Hercules-Witten
	Formaldehyde	Methanol based
	Methyl chloride	Methanol based
Phosgene	2,4-Toluene diisocyanate	Toluene diamine

Propane	Acetaldehyde	
	Acetic acid	
	Acetone	Oxidation of propane
	Formaldehyde	
	MEK, Methanol	
	Propanol	
	Carbon tetrachloride	Propane chlorination
	Perchloroethylene	Propane chlorination
	Nitromethane	
	Nitroethane	Propane nitration
Propylene	Acetone	Wacker
	Acrolein	Propylene based
	Acrylonitrile	Sohio
	Glycerine	Propylene based
	Polypropylene	Various
Propylene oxide	Glycerine	Propylene oxide based
Sulfur trioxide	Alkyl benzene sulfonates	Various
Vinyl chloride	Vinylidene chloride	Acetylene based

Other fluids vaporized in vaporizers

Butanol	Methyl chloride
1-Butene	Methylene chloride
Butylamine	Nitrogen
Bromine	Oxygen
Carbon dioxide	Paracymene
Diethyl ether	Pyridine
Dimethyl ether	Refrigerant 11
Ethane	Refrigerant 12
Ethylene	Refrigerant 13
Hydrogen bromide	Refrigerant 13
Hydrogen fluoride	Refrigerant 223
Hydrogen sulfide	Sulfur dioxide
Isobutylene	Sulfur hexafluoride
LPG mixtures	Turbine fuels
Methylamine	Vinylidene chloride

Design criteria for various fluids. Capacity ratings and flow paths of vaporizers are based upon extensive field measurements of plant scale installations and also on much in-house testing of miscellaneous fluids over a long period of years.

Chlorine. Vertical vaporizers, often with built-in superheat capability, are used. Special instrumentation is required, specifically designed for chlorine service. Base designs are carbon steel. However, if during steamout or cleaning, etc., water is left in the chlorine space, acid is formed that will cause extremely rapid corrosion (hours), often resulting in failure. For this reason, tubes or tubesheets may be used in nickel alloys such as Monel, Inconel 600, and Incoloy 800.

Pamphlet 9 of the Chlorine Institute gives very useful recommendations for application of chlorine vaporizers including some references such as autoignition (rapid corrosion of steel at high temperatures when chlorine encounters a hydrocarbon at the steel surface). Consult the information source about possible difficulties due to concentration of nitrogen trichloride over a long period of time.

Ammonia. Vertical vaporizers, often with built-in superheat capability. Normally steel, but sometimes with stainless steel tubes or tubesheets, particularly if the feed material may have trace elements of a corrosive nature.

TABLE V-5 Atmospheric Boiling Temperatures of Typical Liquids Handled in Information Source's Vaporizers

Fluid	°F	°C
Ammonia	–28.0	–33.3
Chorine	–30.1	–34.5
Hydrogen chloride	–121.0	–85.0
Hydrogen sulfide	–75.3	–59.6
Sulfur dioxide	14.0	–10.0
Methane	–258.7	–161.5
Ethane	–127.8	–88.8
Propane	–43.7	–42.1
N-Butane	31.1	–0.5
Ethylene	–154.7	–103.7
Propylene	–53.9	–47.7

Miscellaneous liquids. Methanol, sulfur dioxide, refrigerants, random hydrocarbon mixtures, etc. Mostly handled in vertical vaporizers, often with built-in superheat capability. *Metals as required to suit individual needs*. Typical metals include Hastelloy, various stainless steels, Carpenter 20, Incoloy 800, Monel, Inconel 600, and various nickel alloys. Low temperature nickel steels are seldom used because of unavailability of small quantities of metal on short notice, plus costly fabricating practices.

Liquified petroleum (propane, butane). Normally in vertical units, sometimes in indirect water bath vaporizers. Normally all steel equipment.

Lethal fluids. Such as hydrogen sulfide, phosgene, hydrogen cyanide, xylol bromide, etc. Normally in vertical units, can be all welded (no gaskets anywhere) if preferred. Usually 100 percent radiographically inspected and heat treated after fabrication. Also available as heavy duty shell design with removable bundles.

Ethylene. Usually fed at low temperature to horizontal shell and coil type vaporizers at very low temperatures (–155°F or –104°C) so stainless steel or other high impact value material is normally used. Typical designs include steel shells with stainless steel vaporizing bundle designed to avoid surging and withstand thermal shock conditions.

Liquefied natural gas. Usually substantially methane, handled like ethylene. Sometimes vertical installations are required for shipboard application, involving approval for such codes as ABS, United States Coast Guard, Lloyds, Veritas, etc.

Cryogenics. Nitrogen, oxygen, low boiling hydrocarbons, etc. Similar to ethylene except sometimes temperatures for direct steam heating may be as low as –325°F (–198°C). Most materials are stainlesses. Special cleaning may be required for oxygen processes.

Freezeup protection. For details, see discussion later in this section. (See Table V-5 for atmospheric boiling temperatures of some typical liquids.)

Cryogenic vaporizers (direct steam heated)

Uses of cryogenic vaporizers. These are used on vaporizing process upset fluids such as ethylene, propylene, etc., on flare systems, where quantities exceed normal capacity of flare drums, or to vaporize ethylene, nitrogen, etc., for consumption out of atmospheric storage systems. They all also used on LNG tankers to vaporize nitrogen, for padding, for loading or transfer of cargo, or in areas between tanks to reduce explosion hazard.

See Figs. V-21 and V-22.

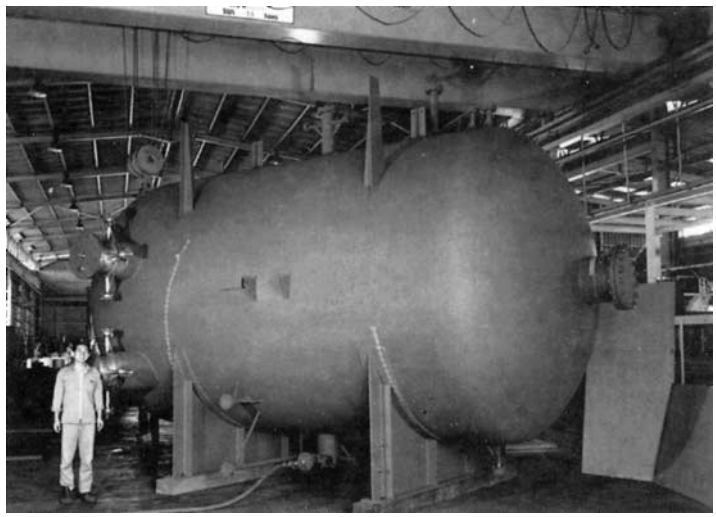


FIG. V-21 Large mixed hydrocarbon vaporizer for feed temperature approximately −150°F (−101°C) for installation in Eastern Gulf refinery. Unit size 132 in (3353 mm) in diameter × 23 ft (7000 mm) long. (Source: Armstrong Engineering Associates.)

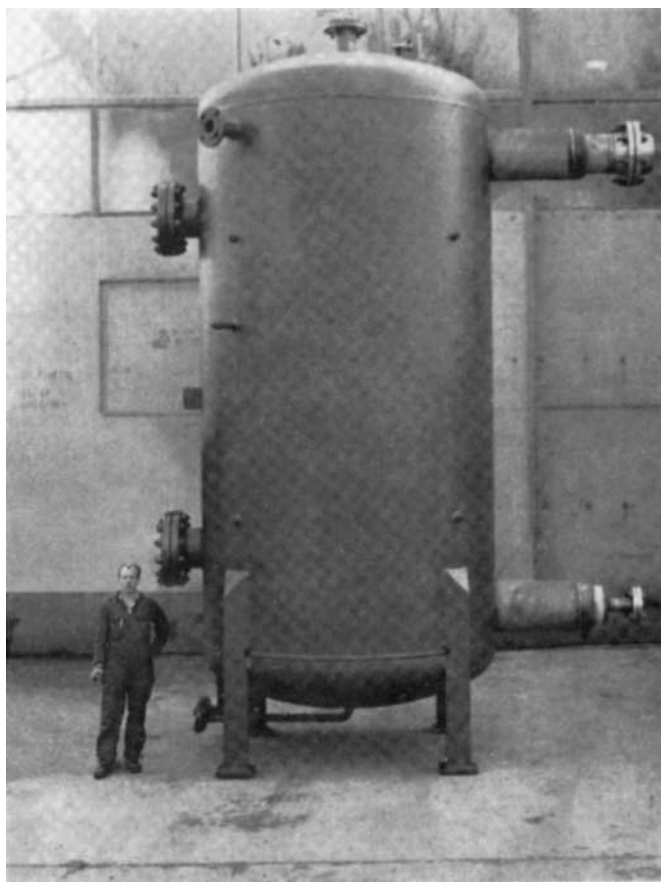


FIG. V-22 Vertical vaporizer/superheater with internal helical coil. Shell is steel with internals of stainless steel. 1412-kW (1,215,000-kcal/h) unit installed on LNG tanker to heat product and assist pump transfer from hold. Vertical format reduces footprint when necessary. (Source: Armstrong Engineering Associates.)

Direct steam heated vaporizers for fluids boiling as low as –327°F (–200°C). Ultra-low temperature vaporizers are designed to avoid freezeup of steam condensate since the steam is on the shellside and condensate is always in contact with the steam.

Indirect systems often require relatively larger equipment and much more costly instrumentation plus the maintenance and supervision that goes with that instrumentation. Direct heated vaporizers require simpler controls than indirect heaters plus avoid any danger of condensate freezeup except if the steam trap gets blocked and traps condensate inside the shell. Desuperheaters can be added to give maximum flexibility to exit vapor control temperature.

Direct heated cryogenic vaporizers have a long history of successful field experience, using direct steam as a heating medium to boil and superheat fluids from as low as –327°F (–200°C). None of these many direct-heated vaporizers, which are in a variety of fluid duties, has ever failed from freezeup of steam condensate to our knowledge.

Vaporizers have considerable antifatigue designs built in for grid loadout duties where the vaporizer may operate on highly fluctuating/cyclic flow rates.

The design and method is thoroughly proven from a great many field installations of sizes up to 40,000,000 Btu/h (10,000,000 kcal/h) per individual vaporizer.

Transfer heaters for very cold liquids antifreeze designs

Typical duties

1. Heating of liquid ammonia, ethylene, LPG, etc., for transfer from ship or other low-temperature storage to ordinary steel pipelines or shipping tanks.
2. Defrosting of fluid circulating systems.
3. Vaporizing oxygen, nitrogen, etc., at temperatures down to –327°F (–200°C) and pressures up to 6000 psi (422 kg/cm^2) using steam or hot water as a heating medium.
4. Heating of very corrosive fluids in separate tube bundles of metal such as Monel, Nickel 200, Inconel 600, Incoloy 800, silicon bronze, etc.
5. Very high-pressure liquid flow can be up to 10,000 psi (703 kg/cm^2 or 680 atm).

Often it is convenient to use a combined two bundle unit in a single shell, or two separate shells using steam to vaporize an intermediate fluid, for example methanol. The vaporized methanol then rises to the top bundle, heating up the fluid passing through the tubes. Since methanol is not subject to freezing, and the boiling temperature of the methanol is kept above 32°F (0°C), there is no freezeup hazard. See Figs. V-23 and V-24.

Small electric indirectly heated vaporizers

Electrically indirect heated vaporizers are suitable for boiling of ammonia and a number of other fluids. Shells are usually of steel, and the heating elements are often copper, although many other metals, such as stainless steels, Monel, Inconel, Hastelloy, and Incoloy can be supplied when requested.

These vaporizers can be supplied with or without controls. They offer a solution to vaporization in outlying areas where steam is not available.

Typical duties include vaporizing HF, H_2S, bromine, CO_2, SO_2, CH_3Cl, Cl_2, HCl, NH_3, LPG, C_3H_8, C_4H_{10}, etc.

See Figs. V-25 through V-28.

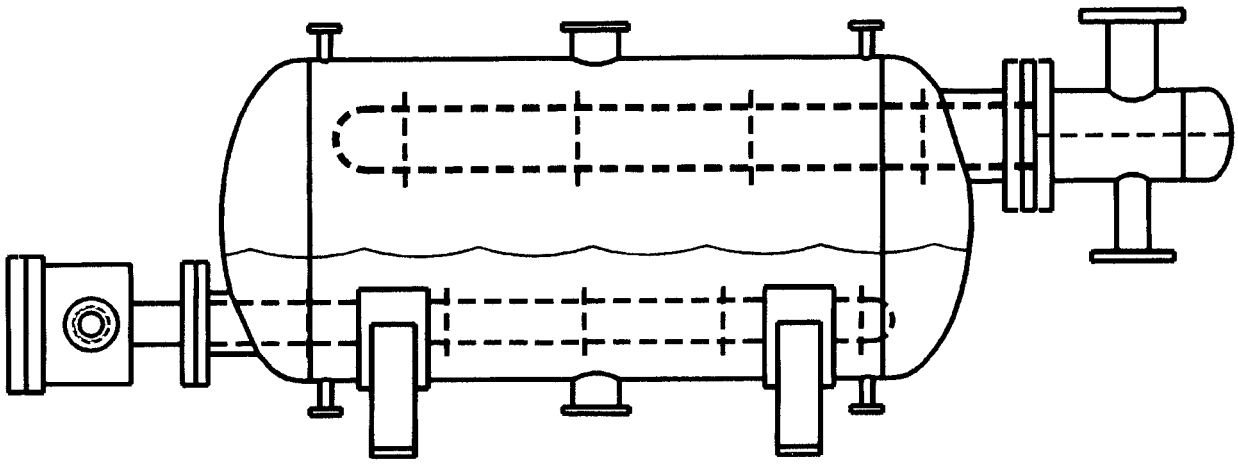


FIG. V-23 Intermediate fluid transfer antifreeze heater. (Source: Armstrong Engineering Associates.)

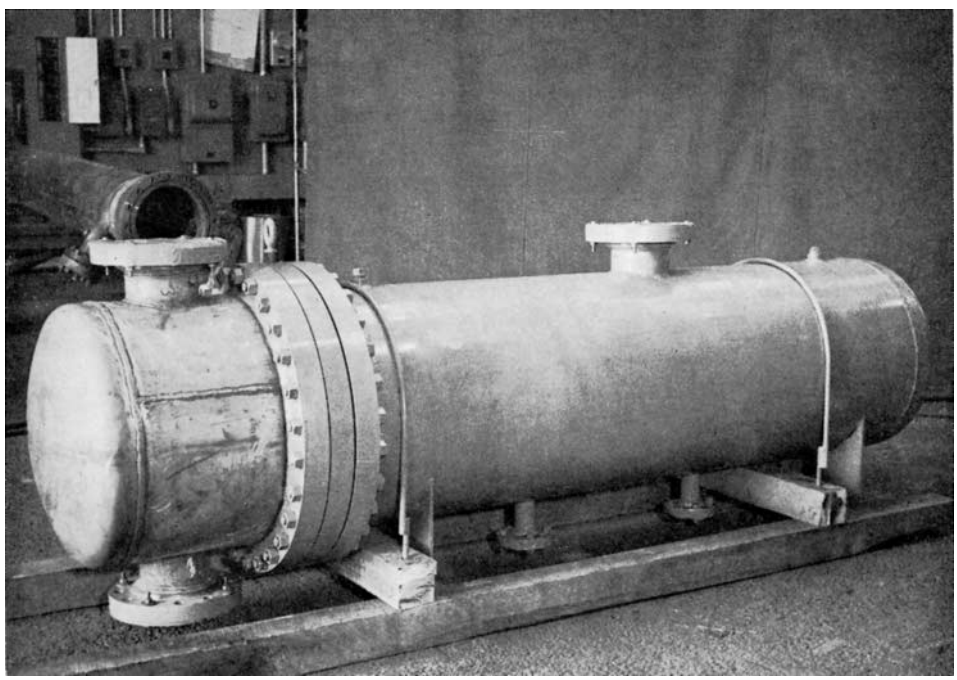


FIG. V-24 Direct heated transfer heater. Horizontal antifreeze heater for cold ammonia, entering temperature −28°F (−33°C) (or below if necessary). Steam heated. Steel and stainless steel construction. Removable bundle. Can also be electric heated. (Source: Armstrong Engineering Associates.)

Combination electric indirect vaporizers

In working with highly inflammable or explosive fluids, it may not be safe to put the electric element directly in contact with the fluid. Therefore, the information source supplies a combination unit consisting of a vaporizer, an electric water heater, and a pump, with controls.

The electrically heated water is circulated into the vaporizer to furnish the heat required for boiling. This same arrangement may be used for chlorine, or for any

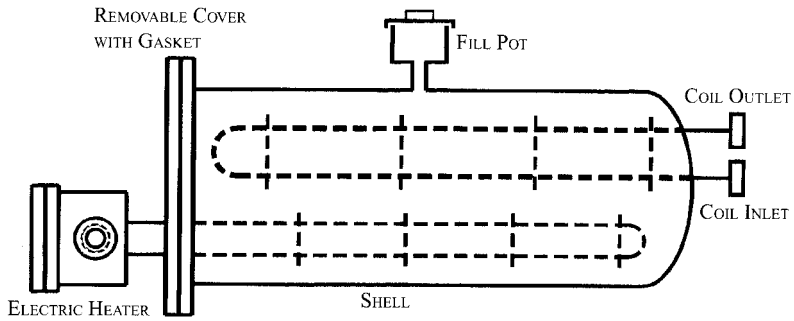


FIG. V-25 Typical arrangement for electric water heater skid mounted unit to vaporize whatever fluid desired. (Source: Armstrong Engineering Associates.)

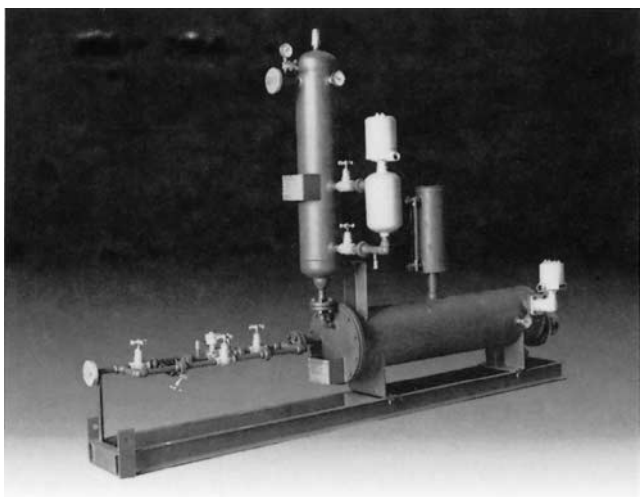


FIG. V-26 Electric indirect heated (glycol bath) vaporizer, for boiling of propane, LPG, HCl, SO_2, etc. (Source: Armstrong Engineering Associates.)

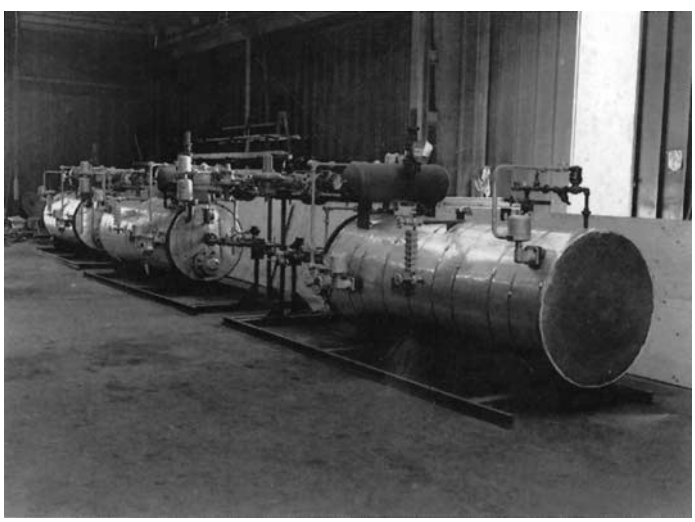


FIG. V-27 Line of 36 in (914 mm) × 8 in (2438 mm) indirect heated LPG vaporizers showing insulation and controls. Installed in large apartment complex in Hong Kong. (Source: Armstrong Engineering Associates.)

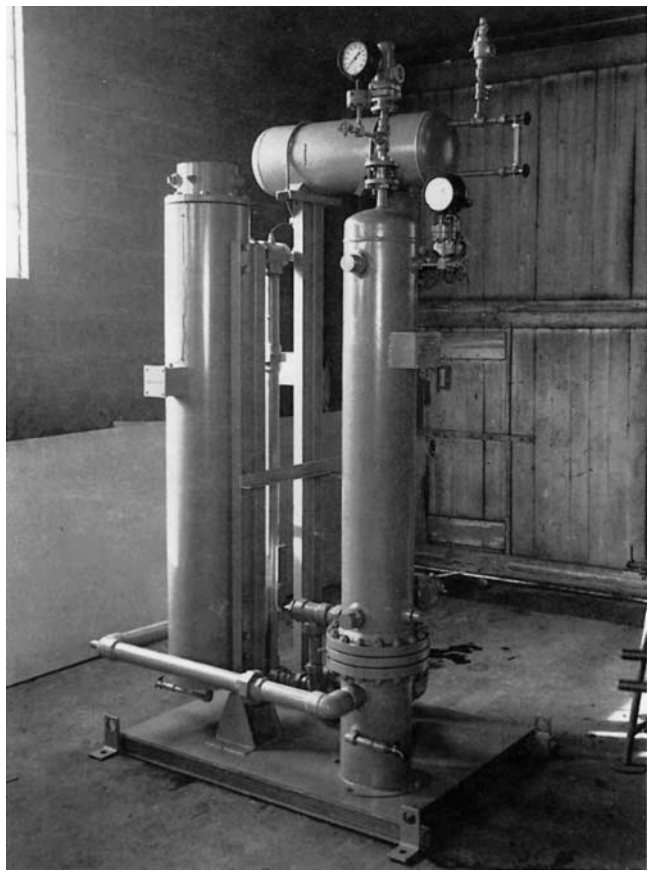


FIG. V-28 Electric heated vaporizer: Circulating pumped water bath. System includes electric immersion water heater, pump, piping, controls, and expansion tank, all factory piped up and delivered on skid ready to operate. (Source: Armstrong Engineering Associates.)

fluid that might offer severe corrosive attack to the sheath metal of the electric heater. This typed unit is often used for C_3H_8, Cl_2, C_4H_{10}, SO_2, Freons, etc.

See Figs. V-29 and V-30.

Pressurizing storage tanks

Where the outdoor storage temperature is very low, the saturation pressure of the stored liquid may get so low there is not sufficient pressure to deliver the liquid or vapor past the piping and valve resistances required. In such cases, a vaporizer may be used to furnish the heat required to keep the stored liquid at a desired temperature and corresponding pressure, even though the outdoor ambient temperature may be quite low. See Fig. V-31.

The most prevalent type of vaporizer for this purpose is the vertical bayonet. The main point to consider is whether there is a likelihood of damming or stoppage of the condensate inside the tube at a time when surrounding liquid is below 32°F (0°C). In that event, freezing is a danger and in such times, the antifreeze aspect of the vertical bayonet vaporizer becomes desirable.

The internal condensing capacity of the storage tank, even with only a fraction of the surface not covered by liquid, is tremendous, and would require both a very large vaporizer and also a very much larger boiler or heat source. Therefore, the only practicable way to approach this problem is to figure on boiling the liquid at

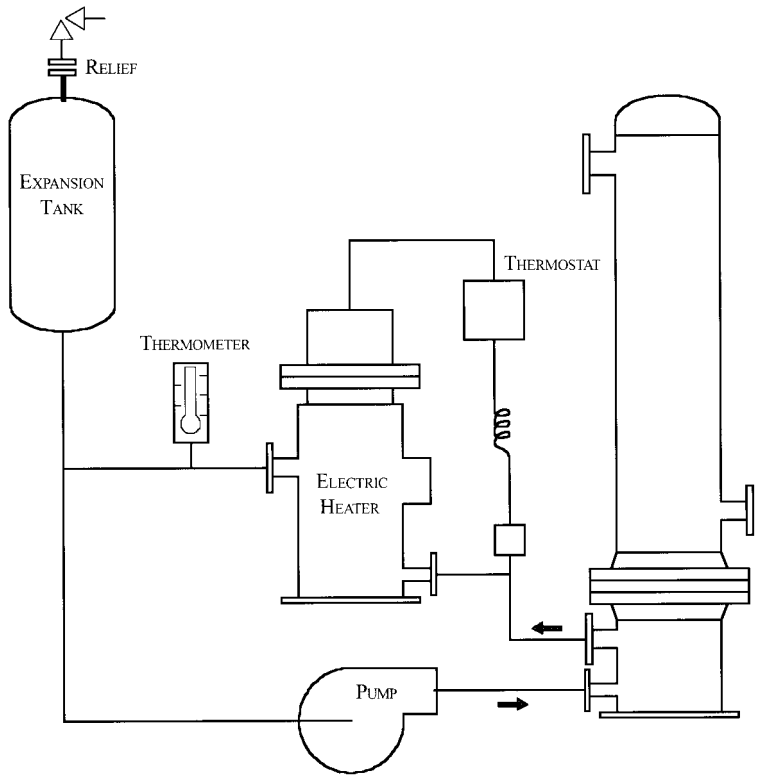


FIG. V-29 Electric indirect vaporizer in process and flow. (Source: Armstrong Engineering Associates.)

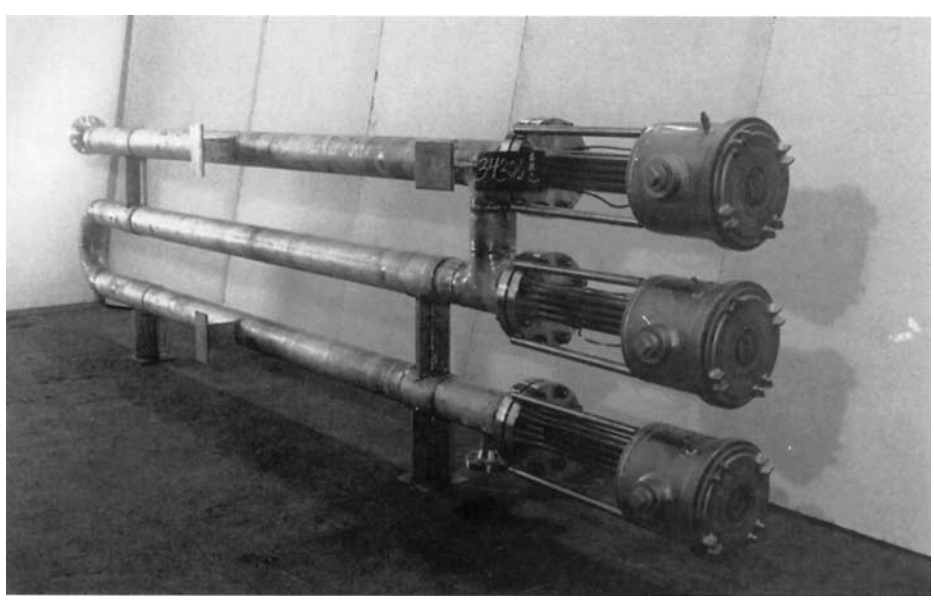


FIG. V-30 Direct electrically heated ammonia vaporizer. (Source: Armstrong Engineering Associates.)

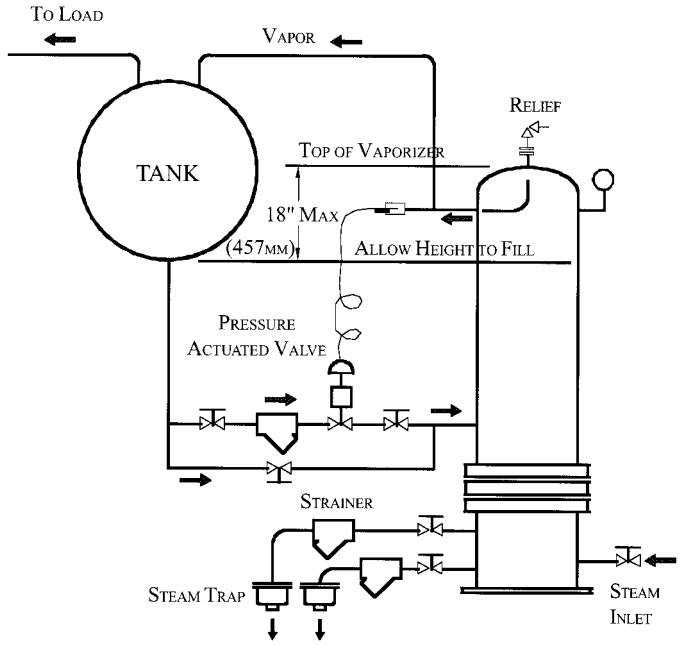


FIG. V-31 The above vaporizer hookup shows an arrangement that is often used for ammonia and other gases in addition to propane. The liquid level is important to make sure the liquid will flow by gravity or otherwise into the vaporizer. The pipelines must be large enough to overcome any hydraulic loss in the flow system, to make sure that the vaporizer tube bundle will be covered. Otherwise, at low levels, the vaporizer will not have full capacity. The pressure-actuated valve may not be absolutely necessary, but is desirable in many cases. Since it is usually better to keep the steam on at all times, this avoids excess boiling at times when the tank may already have adequate pressure. (Source: Armstrong Engineering Associates.)

the beginning of operation at the low temperature; get the whole body of the tank and its stored liquid up to the operating temperature, say 70°F (21°C), and then furnish enough heat to overcome the convection heat loss from the outside surface of the tank to the ambient air. Also, sufficient capacity is needed to heat the fresh incoming liquid as it arrives by tank car, assuming that it also has cooled down to the outdoor temperature.

To illustrate the application, take an example of a 30,000 U.S. gallon uninsulated storage tank, 9 ft in diameter by 63 ft long, to be filled with 10,000 U.S. gallons of liquid anhydrous ammonia in a 10-h day, assuming 70°F required tank temperature, and with an ambient outdoor temperature of –30°F. The outside surface area of the tank is 1827 ft^2. The convection heat loss can be taken conservatively as 5 Btu/h/ft^2/°F at a wind speed of 20 mph.

Ambient heat loss is then

$$1827 \times 5 \times [70 - (-30)] = 915{,}000 \text{ Btu/h}$$

Incoming fresh liquid (loading the tank):

$$1000 \text{ gph} \times 5.6 \text{ lb/U.S. gal} \times (120.5 - 10.7) = 615{,}000 \text{ Btu/h}$$

Note: 120.5 is enthalpy of liquid at 70°F (21°C); 10.7 is enthalpy of liquid at –30°F (–34°C).

Ambient heat load	913,500 Btu/h
Fresh liquid heatup	615,000 Btu/h
Total	1,528,500 Btu/h

To illustrate another application, we take an example of a 114-m^3 uninsulated storage tank, 2.74 m in diameter by 19.2 m long, to be filled with 38 m^3 of liquid anhydrous ammonia in a 10-h day, assuming 21°C required tank temperature (8.0 kg/cm^2G required pressure), and with an ambient temperature of –34°C. The outside surface area of the tank is 170 m^2. The convection heat loss can be taken conservatively as 24.4 kcal/h/m^2/°C at a wind speed of 8.9 m/s. Ambient heat loss is then

$$(170) \times (24.4) \times [(21) - (-34)] = 228{,}140 \text{ kcal/h}$$

Incoming fresh liquid (loading the tank in 10 h):

$$3.8 \text{ m}^3 \times 683.17 \text{ kg/m}^3 \times (66.94 - 5.94) = 158{,}360 \text{ kcal/h}$$

Note: 66.94 kcal/kg is enthalpy of liquid ammonia at 21°C; 5.94 kcal/kg is enthalpy of liquid ammonia at –34°C.

Ambient heat load	228,140 kcal/h
Fresh liquid heatup	158,360 kcal/h
Total	386,500 kcal/h

This duty is added to the amount of vapor withdrawn from the tank to process, if any.

Vaporizers with controls

Setting of float valves on vaporizers. The fluid that is boiling inside the vaporizer is not entirely liquid, but a mixture of liquid and vapor. As a result, its density is usually appreciably lower than the density of the liquid column, which acts upon the float in the float chamber, which is not subject to such vigorous boiling. See Figs. V-32 through V-39.

Therefore, the height at which the operating center of gravity of the float should be set should be lower than the actual boiling level in the vaporizer by an amount equal to the ratios of the densities of the liquids in the vaporizer to that in the float column. A good practice is to start at about 2/3 the height of the vaporizer level above its bottom, and then adjust the float level in the field to give best performance. Note that on vertical vaporizers, the float acts only as a limit to capacity. When the vaporizer is operating at less than its capacity, it automatically operates with a liquid level lower than the top of the tubes.

On all installations, if the propane or other liquid in the shell should get down to temperatures below 32°F (0°C), it is very important not to have a steam fallure. In this case, the vapor can condense and freeze and possibly burst a tube.

This is important to consider in any control that will shut the steam off and also on any system where the steam may fail for external reasons.

When desirable, vaporizers can be supplied with the controls mounted. We caution the prospective buyer in this case:

1. Since there is wide variety in preferences as to method, make, and type of control, it often takes several times as long to work up quotations on units with controls as it does to quote bare vaporizers.

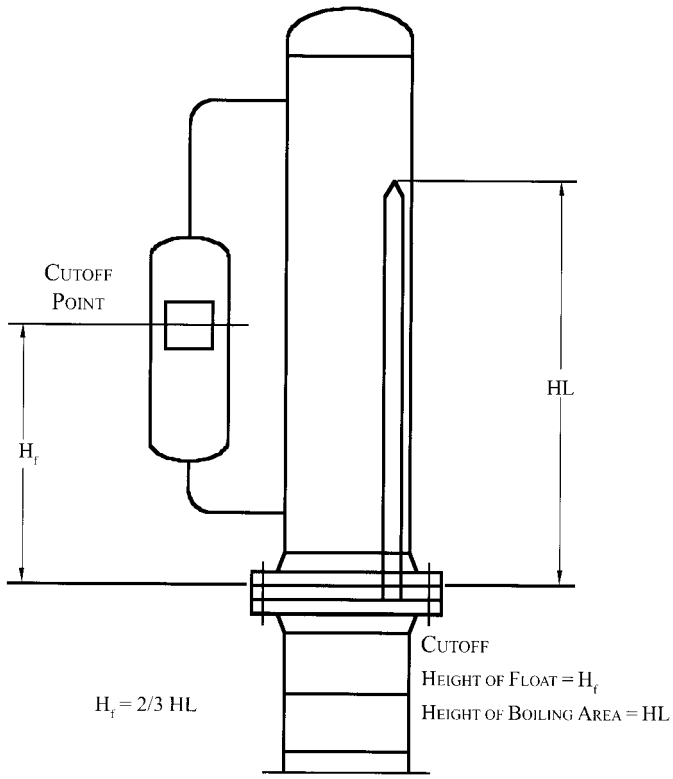


FIG. V-32 Float setting. (Source: Armstrong Engineering Associates.)

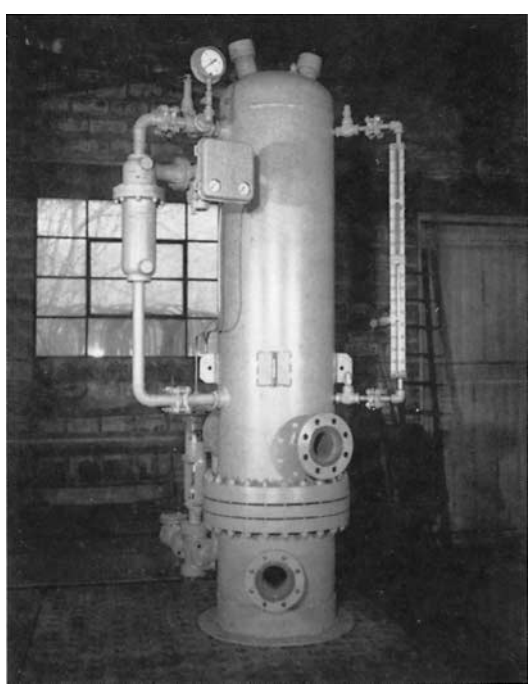


FIG. V-33 Process vaporizer with controls factory mounted. Note float and float-operated valve, liquid level gauge, thermostatic steam valve, condensate traps, bursting disc relief valves, and miscellaneous hand valves, including bypasses and strainers. (Source: Armstrong Engineering Associates.)

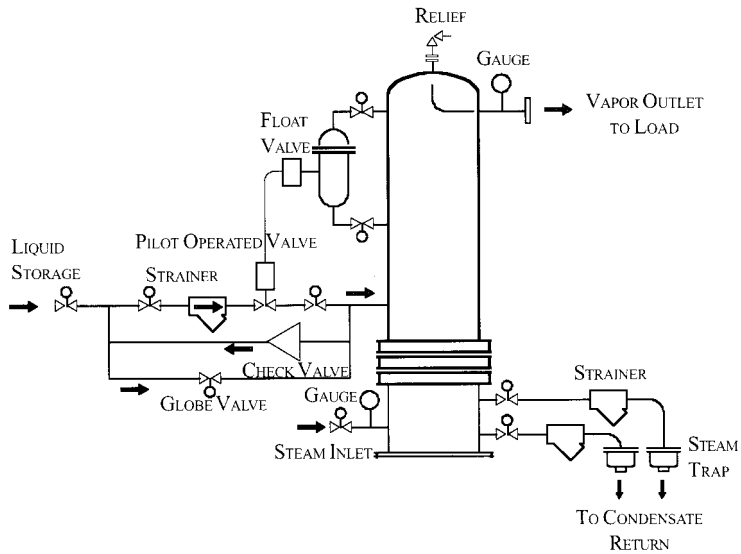


FIG. V-34 Typical control hookup for routine fluid vaporizing. Note presence of shellside float valve. This operates to stop liquid feed when flow level gets too high due to excessive draw of fluid from the vaporizer. The valve shuts down flow of the liquid feed to the vaporizer to avoid carryover of liquid into the exit line. (Source: Armstrong Engineering Associates.)

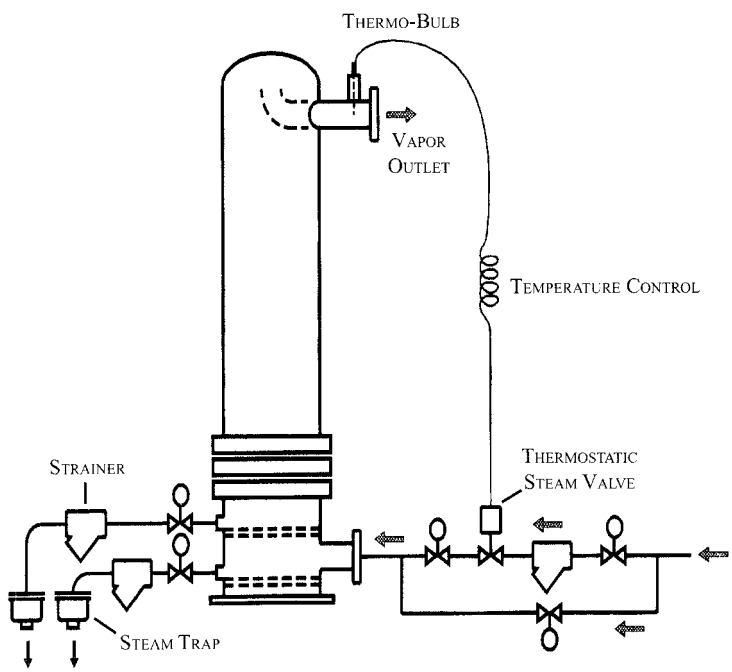


FIG. V-35 Typical control setup when specific superheated temperature is required. For normal chlorine vaporizing see separate diagram. (Source: Armstrong Engineering Associates.)

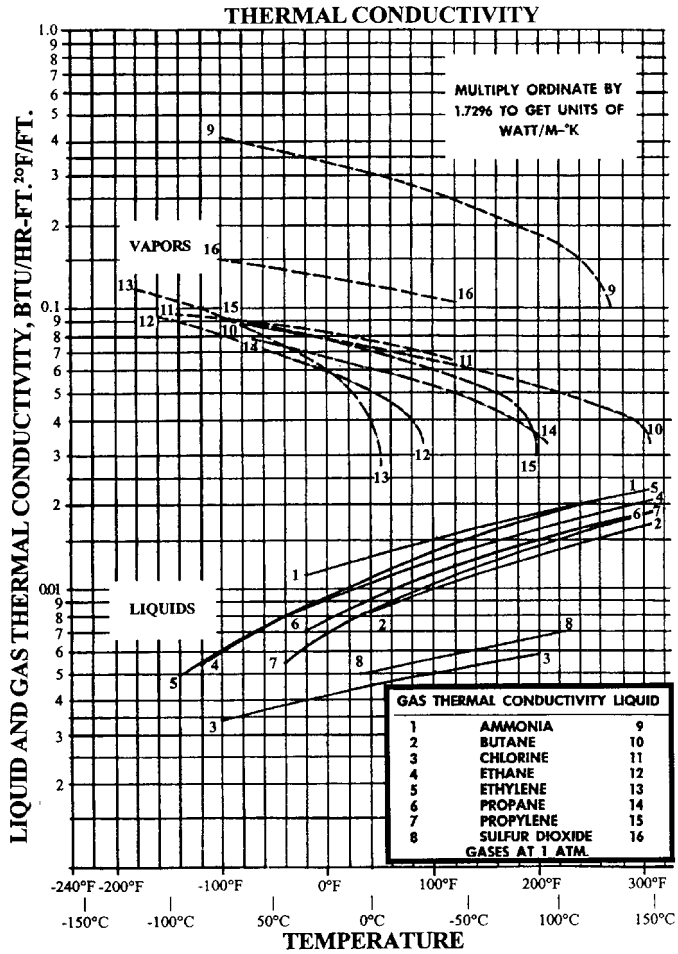


FIG. V-36 Conductivity—temperature for various fluids. (Source: Armstrong Engineering Associates.)

2. It usually takes several weeks longer to supply vaporizers with controls than it does to complete a bare vaporizer.

Liquid to vaporizer–vapor direct to load (vapor does not return to storage tank). Figure V-34 shows the most common dependable hookup for automatic operation. Liquid from the tank enters either by gravity, differential pressure, or may be fed by liquid pump when necessary to ensure adequate pressure to go through the reducing valves. This liquid enters the three lines and may go alternatively through the valve under normal operation, or through the hand bypass if the pilot valve is out. The pilot valve is controlled by a float on the shell that limits the level to which the liquid can rise above the tubesheet face.

There may or may not be a pressure-reducing valve between the storage tank and the controls shown in the figure, and depending on the duty, there may be a further vapor pressure-reducing valve after the outlet, before the load.

In most cases, the steam is fed by hand feed only, and is left on all the time. A trap may be used to drain the regular condensate outlet, and another one to also drain the condensate that forms in the steam chamber.

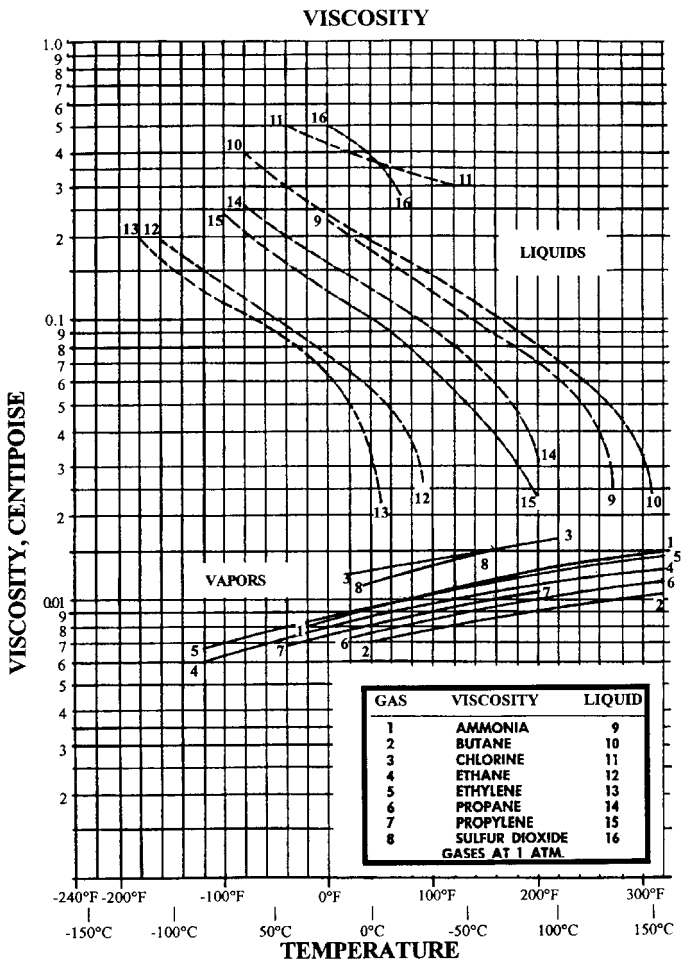


FIG. V-37 Viscosity temperature for various fluids. (Source: Armstrong Engineering Associates.)

If the unit is operating normally and the vapor outlet line is suddenly closed due to load variations, the steam being on, the pressure would normally build up rapidly with the liquid being confined in the vaporizer. This might cause the relief valve to blow and lose fluid. To avoid this, the center bypass check is installed—called a "blowback." When the vaporizer pressure exceeds the feed pressure, the flow will reverse, allowing the liquid to leave the vaporizer and return to the tank. Obviously, this blowback line must take into consideration the particular design of pump and pressure relief valves, if either are present, and must be arranged to bypass them if they will not permit reverse flow.

Constant vapor outlet temperature. The hookup illustrated in Fig. V-35 shows an alternate arrangement that is sometimes used to give a fairly constant vapor outlet temperature. It may be combined with a constant pressure arrangement through use of pumps and reducing valves if the pressure must also be controlled to obtain constant gas density for the requirements of mixing valves.

This sketch also shows an alternate arrangement sometimes used to get a constant gas density for an air-gas mixing valve. The thermostatic valve will maintain a constant gas-off temperature and a constant propane pressure in the vaporizer.

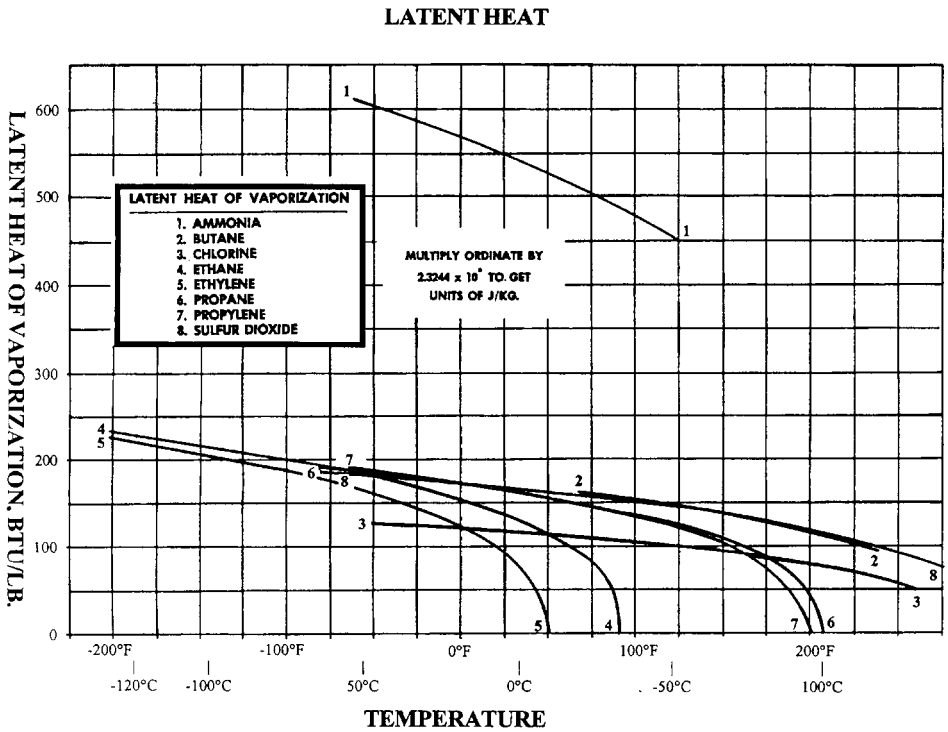


FIG. V-38 Latent heat—temperature for various fluids. (Source: Armstrong Engineering Associates.)

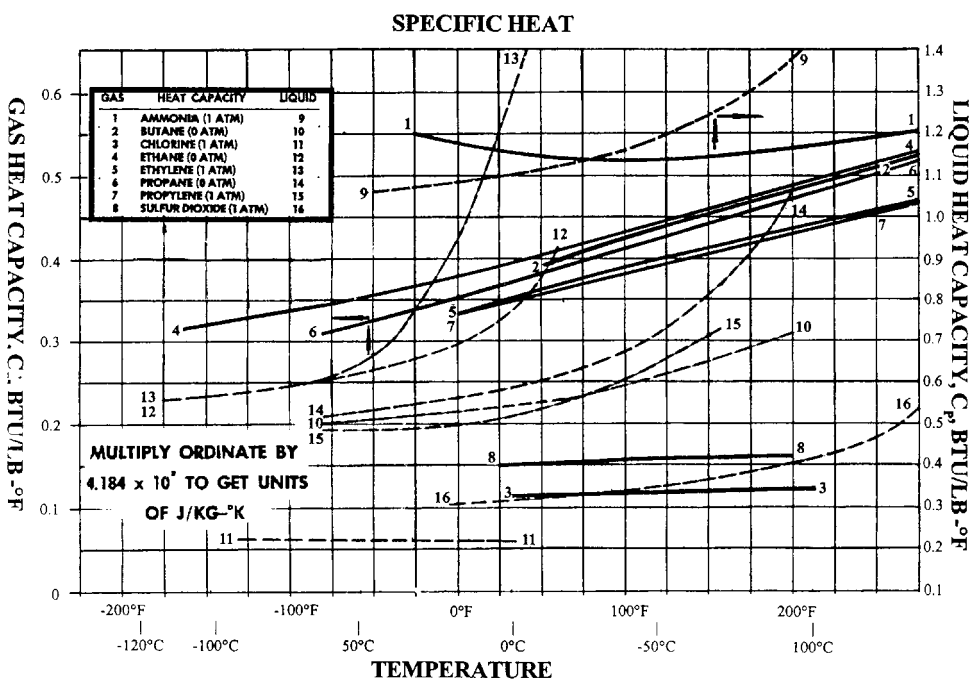


FIG. V-39 Heat capacity—specific heat—temperature for various fluids. (Source: Armstrong Engineering Associates.)

Vaporizers will furnish superheat if operated at lower ratings than those shown. The amount of superheat available by operating at reduced liquid level and using the upper part of the bundle for superheating, however, is limited. If a large amount of superheat is required, then a separate superheater may be advisable from a cost standpoint, as it often takes a great deal more surface to superheat the gas than to vaporize the liquid.

Vents and Flame Arrestors

Vents are provided on a storage tank so that the tank can "breathe." "Inhale" happens if a temperature drop causes a liquid volume reduction. It also can occur if liquid is withdrawn from the tank. "Exhale" happens in the reverse circumstances. Required thermal venting capacity is determined from tank capacity and fluid flashpoint. A tank supplier will ask what fluid your tanks will hold and the steady temperature and pressure at which the fluid is to be held. Vent capacity can then be determined. Conservation vent valves have just a slightly different design to minimize evaporative losses.

Flame arrestors are provided on tanks storing fluids with a flashpoint of less than 110°F. Flame arrestors should also be provided on storage tanks for liquids with a flashpoint above 110°F, where a combustible environment exists or other potentially hazardous condition, such as the heating of heavy oils.

Vibration Measurement (*see* Condition Monitoring)